Soil Cost of Burrowing with run_config_tick4452

Does the burrowing claw drive the prey extinct when burrowing damages the soil? 40 simulations of 150,000 ticks each from the settings in `run_config_tick4452.csv`: 20 with a soil cost of 10 per burrow and 20 with harmless burrowing (cost 0).

Summary

- **Claws took over, then every prey died, in 18 of 20 runs with soil cost 10, and in none of 20 with harmless burrowing** (two-sided Fisher's exact test, $p = 3.4 \times 10^{-9}$). This is the plotting script's *evolutionary suicide* count.
- **Every take-over that had time to play out ended the same way.** With soil cost 10 the claws took over 22 times in total. 18 of those take-overs were followed by every prey dying, a median of 21,000 ticks later (10,000 to 41,000). The other 4 began at tick 110,000 or later, so they had less time than the slowest collapse needed. With harmless burrowing the claws took over in 18 runs and the prey never died out afterwards.
- **With plant regrowth on, the extinction didn't always last.** The world keeps placing new plants, and their descendants can evolve into prey. At tick 150,000, 7 of the 20 soil-cost runs had no prey (5 with nothing alive at all, 2 with only plants), against none with harmless burrowing ($p = 0.008$). In 9 runs a new, almost clawless prey population had grown back, and 4 still had clawed prey after a late take-over.
- **The plants are what crash.** In this configuration a plant needs completely untouched soil (minimum plant fertility 10, the maximum), and a burrowed cell takes about 100,000 ticks on average to recover. With soil cost 10 the plants fell from a median peak of about 630 to a median low of 3; with harmless burrowing their median low was 372.
- **Recovery took most of the run.** In the 7 runs where a large prey population came back, prey kept dying out and reappearing until ticks 102,000 to 131,000, 70,000 to 105,000 ticks after they first died out. That is in line with the roughly 100,000 ticks a burrowed cell needs to recover.

Setup

Every setting came from `run_config_tick4452.csv`, exported from the browser, on its 306 by 140 grid with cell size 5. The headless runner (`analysis/headless/run_sim.js`) uses the branch's own `src/` code, and calls the branch's `WorldEnvironment.regrowPlant` every tick, so plant regrowth works exactly as in the browser. The runner reproduces the export's starting world: 54.3% of the grid is fertile soil (the rest is permanently infertile paths), with the same 10 predators, 20 prey and 39 plants placed.

The two setups differ only in the hide fertility cost. Each was run from 20 starting layouts (placement seeds 1 to 20; the export used seed 1), so every layout was run once with each soil cost. The simulation itself isn't seeded, so a rerun gives similar rates, not identical runs.

Outcomes. Claws *take over* once at least 50 prey have a claw and they make up at least half of all prey. *All prey died out* means the prey count reached zero at a sample (every 1,000 ticks) after a take-over, as in `plot_evo_suicide.py`. Because prey can come back with plant regrowth on, the state of each run at tick 150,000 is reported as well.

Setting	Value	Setting	Value
Hide fertility cost	**10** or **0**	Hide duration	200 ticks
Min plant fertility	10 (= max fertility)	Min burrow fertility	1
Soil regen chance	0.01% per tick, +1	Plant regrowth chance	1% per tick
Starting organisms	10 predators, 20 prey, 39 plants	Starting claws	none (claws must evolve)
Burrow only when close	off (any predator sighting)	Stay hidden near predators	off
Don't age while hidden	on	Claw cells are free	on
Clawed killer becomes predator	0.5 chance	Predator lifespan multiplier	3
Lifespan multiplier	150	Burrower / predator move interval	1 / 3 ticks

How the runs ended

	Soil cost 10	Harmless (cost 0)
Claws took over at least once	20 / 20 (median first at tick 11,000)	18 / 20 (median first at tick 18,000)
All prey died out after a take-over	**18 / 20** (median tick 33,500)	**0 / 20**
State at tick 150,000		
Nothing alive	5 (died at ticks 36,014 to 96,581)	0
Plants only, no prey	2	0
Prey came back, almost no claws	9 (3 to 211 prey, at most 2 with a claw)	0
Clawed prey still alive	4 (1, 4, 38 and 45 prey, all clawed)	18 (78 to 100% of prey have a claw)
Clawless prey, claws never took over	0	2

With harmless burrowing the claw spread in almost every run and stayed: at tick 150,000 the population was around 1,000 (median 976) and plants numbered 464 to 672. Prey did die out in 3 harmless runs (seeds 6, 13 and 17), each time *before* the claws took over, and in seed 6 for about 60,000 ticks. So this food web can crash without any soil damage, but after a claw take-over it only crashed when burrowing damaged the soil.

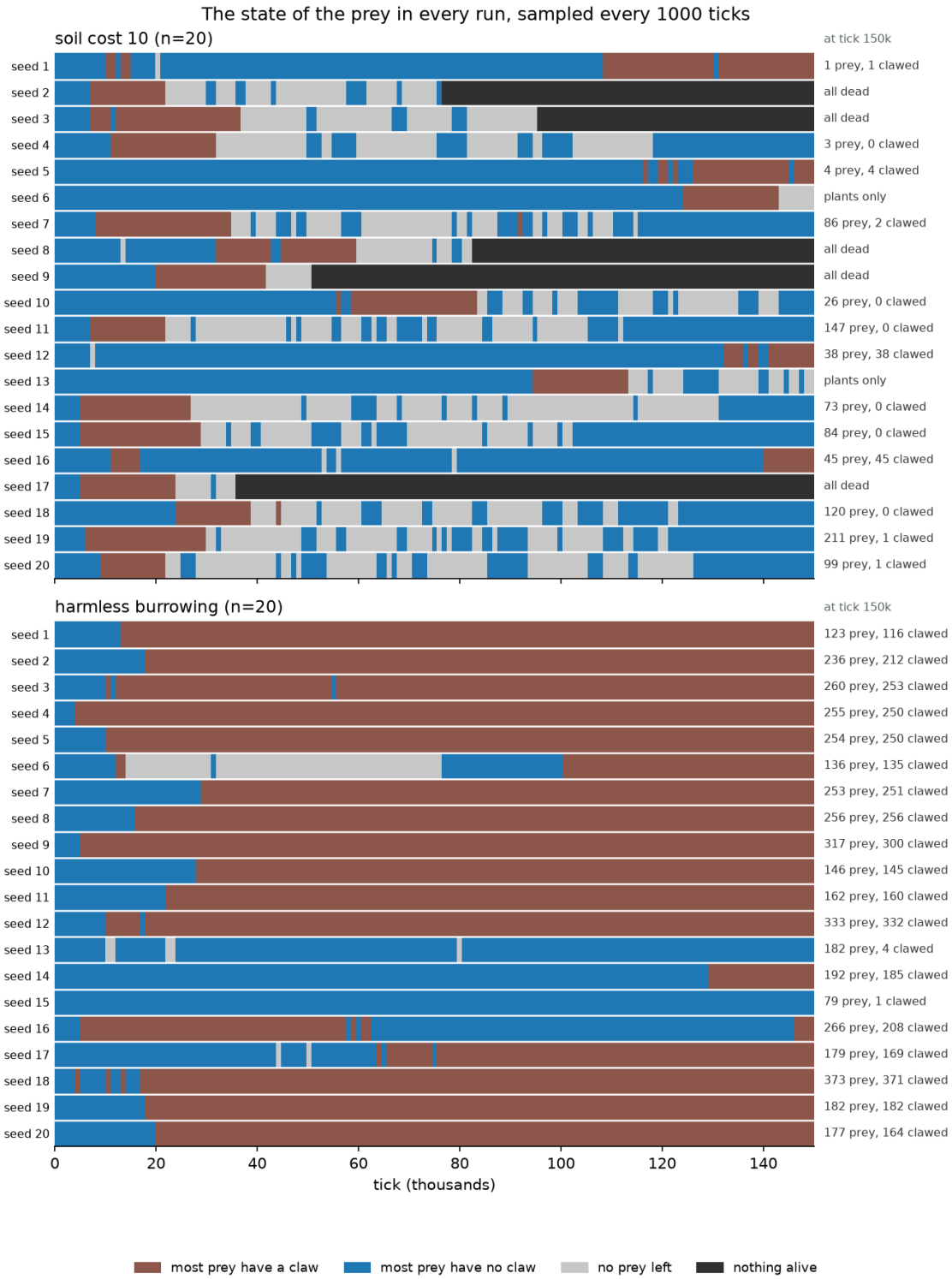


Figure 1. Every run, one row per starting layout, coloured by the state of the prey at each sample. Brown: at least half the prey have a claw. Blue: most prey have no claw. Light grey: no prey alive. Dark grey: nothing alive. The right-hand column gives the prey at tick 150,000. With soil cost 10, nearly every brown stretch ends in grey within a few tens of thousands of ticks; the prey then flicker in and out (blue and grey) until the soil recovers. With harmless burrowing, the brown stretches last to the end of the run.

What happens after the claws take over

Figure 2 lines every run up on the tick the claws took over. With soil cost 10 the clawed prey peak within a few thousand ticks, the average fertility of the fertile soil falls to about 60% of its maximum (median low 58.7%, range 51 to 83.5%), and the plants drop from around 500 to almost none within about 15,000 ticks. The prey run out of food and die, and the predators with them: predators died out in 19 of the 20 runs. With harmless burrowing the soil is untouched and the plants and clawed prey hold steady.

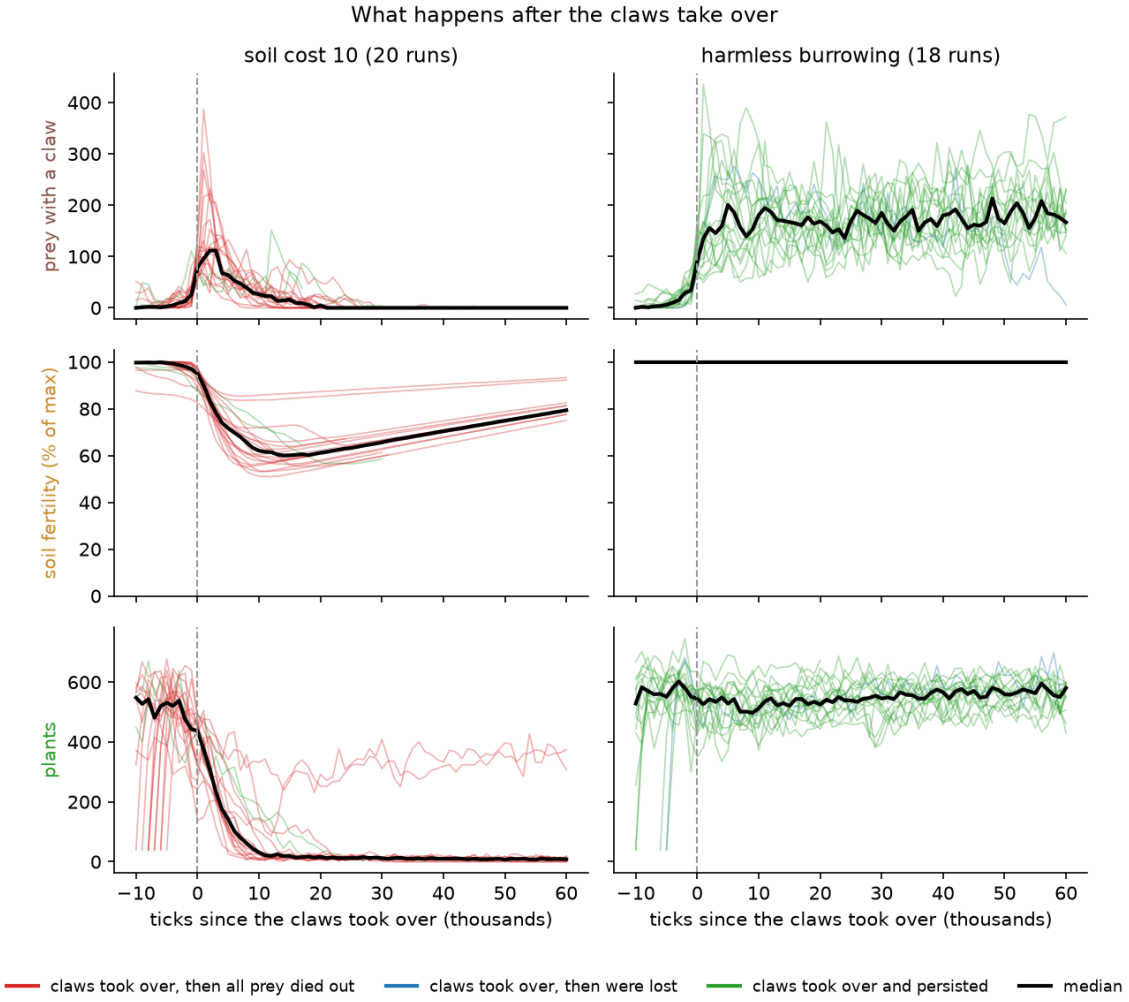


Figure 2. Every run lined up on the tick the claws first took over (dashed line), median in black. Top: prey with a claw. Middle: mean fertility of the fertile soil, as a percentage of its maximum. Bottom: plants. Red: counted as *all prey died out*. Green: the claws took over and persisted. Blue: the claws took over and were later lost.

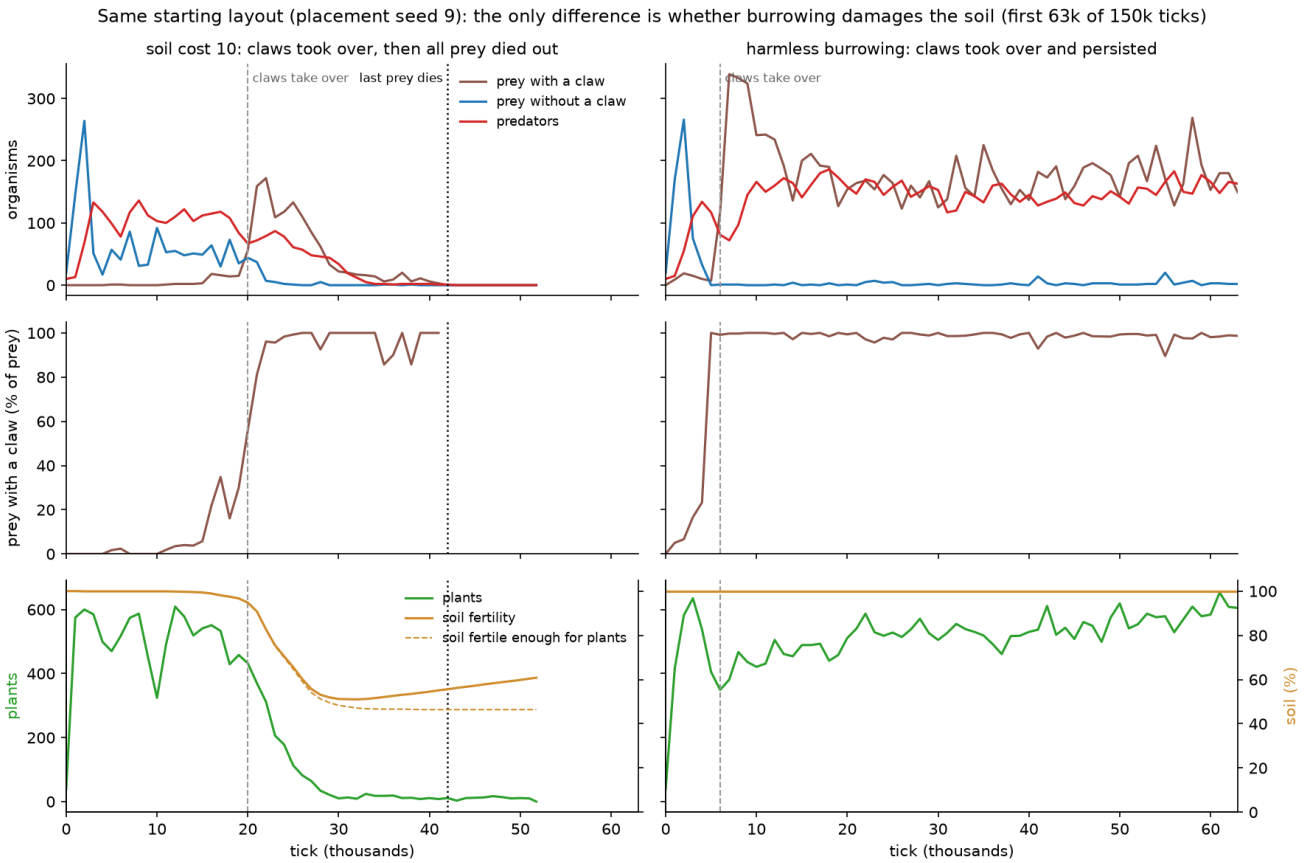


Figure 3. One starting layout (seed 9) run with each soil cost. With soil cost 10 the claws take over at tick 20,000, the soil fertile enough for plants (dashed) falls below half, the plants die off and the last prey dies at tick 42,000; nothing is alive after tick 51,763. With harmless burrowing the claws take over at tick 6,000 and both clawed prey and predators are still thriving.

Why the plants crash

- **A burrow empties the soil.** Hiding removes 10 fertility from every soil cell under the organism, and 10 is the maximum, so each burrow leaves a patch at zero.
- **Plants need perfect soil.** A plant can only be born or placed where the average fertility under it is at least the minimum plant fertility, 10 here, so every cell under it must be at full fertility. A single burrow makes a patch unusable for plants.
- **Soil recovers very slowly.** An empty cell regains 1 fertility with a 0.01% chance per tick, about once every 10,000 ticks, so a cell takes about 100,000 ticks on average to recover from 0 to 10.
- **The burrowing happens where the plants are.** Prey burrow wherever they are when they see a predator, which is likely to be among the plants they feed on. As those plants die they can't be replaced, and the share of fertile soil good enough for plants fell to a median low of 54% (46 to 81%).

Recovery through plant regrowth

Plant regrowth places a new plant on good soil about once every 100 ticks (about 1,500 per run), and any descendant that evolves to move counts as prey. So after the prey died out, new prey kept appearing and dying out again: the 7 runs that ended with a large prey population (seeds 7, 11, 14, 15, 18, 19 and 20, with 73 to 211 prey) recorded 8 to 13 separate die-outs each, the last between ticks 102,000 and 131,000. By the end, 90 to 95% of their fertile soil was good enough for plants again, the plants were back at 353 to 524, and predators had returned (26 to 111). Few of these new prey had claws: none of the 7 runs had more than 14 clawed prey after their last die-out.

Claws can evolve again. In seeds 1 and 16 the claws took over a second time, at ticks 110,000 and 143,000, after the first take-over had wiped out the prey, and both runs still had clawed prey at the end. In seeds 6 and 13 the

claws first took over late (ticks 126,000 and 97,000) and the prey had died out by tick 150,000. Measured at tick 100,000 instead of 150,000, 10 of the 20 soil-cost runs had no prey.

In the browser's chart

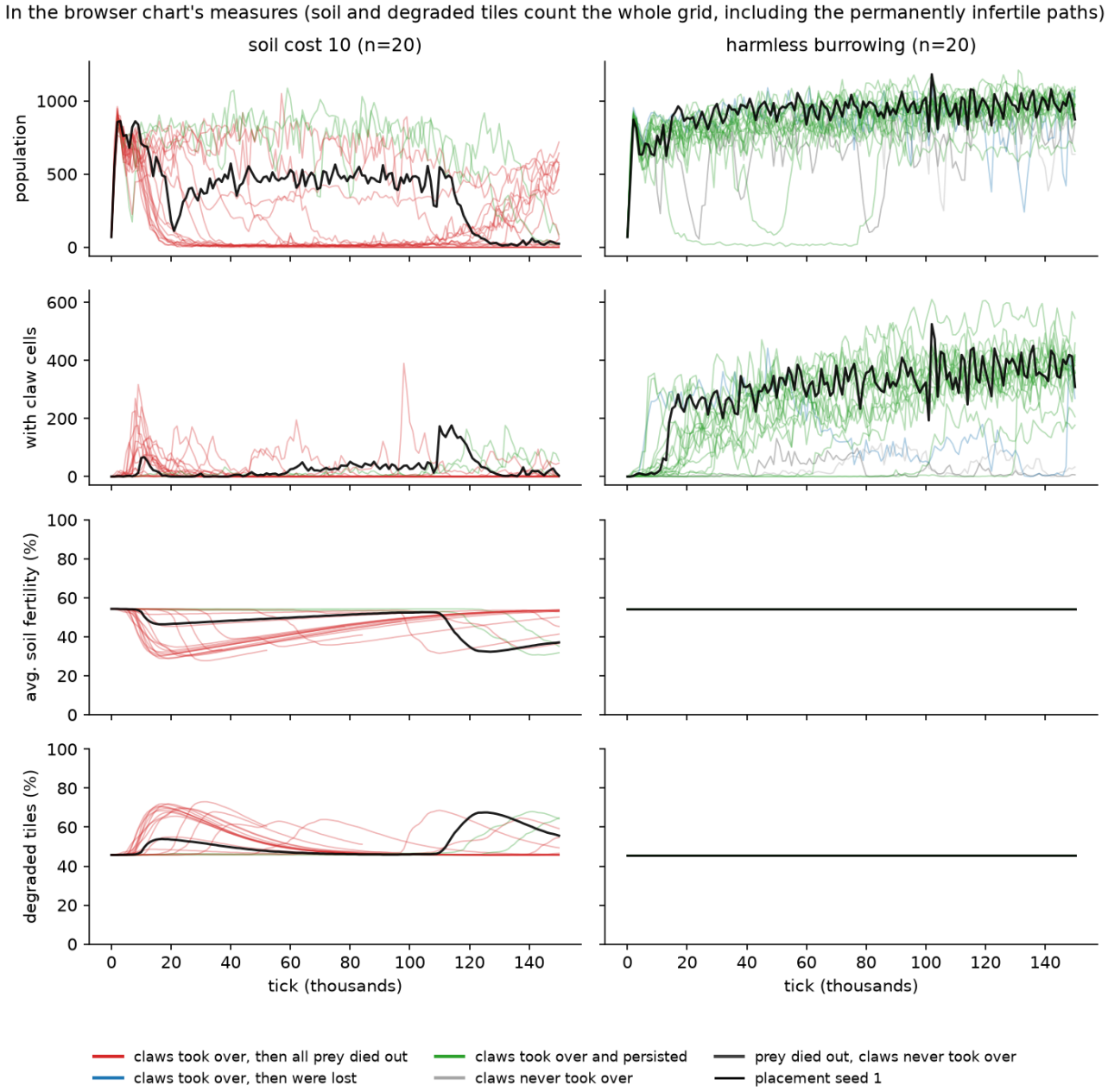


Figure 4. The same runs in the measures of the browser's *Burrowing & Soil Health* chart, for comparison with browser runs. Placement seed 1, the export's layout, is in black. Colours as in Figure 2.

The browser's soil line averages over the whole grid, including the infertile paths that cover 46% of this grid, so it starts at 54.3% and can't go higher. With soil cost 10 it fell to a median low of 31.8%, and degraded tiles (below half the starting average) rose from 45.7% to a median peak of 68.2%; with harmless burrowing both lines stay flat. In these units a crash shows as the population falling to near zero while the soil line dips and then slowly climbs back. The soil recovering on the chart comes after the prey have gone, so it isn't a rescue.

Statistics

Comparison (20 runs each)	Soil cost 10	Harmless	Fisher's exact p (two-sided)
All prey died out after a take-over	18	0	3.4×10^{-9}
No prey at tick 150,000	7	0	0.008
Nothing alive at tick 150,000	5	0	0.047

Limitations

- These are headless runs of the branch's code. The code is the same as in the browser, but the rates haven't been checked against browser runs.
- 20 runs per setup, all on one soil layout (soil seed 42 on the 306 by 140 grid); only the placement of the starting organisms varied.
- Populations were sampled every 1,000 ticks, so a die-out shorter than that can be missed, and the number of separate die-outs is approximate.
- The runs stop at tick 150,000. The 4 take-overs from tick 110,000 on hadn't had the 10,000 to 41,000 ticks that the earlier take-overs took to wipe out the prey, so their outcome is open.
- This configuration differs from the earlier `run_config_tick73074.csv` sweep (4 of 20 runs with all prey dying out over 100,000 ticks) in several ways at once: minimum plant fertility 10 instead of 7, staying hidden near predators off, predator lifespan multiplier 3 instead of 2, burrowers moving every tick, and 150,000 ticks. The perfect-soil rule for plants is the likely driver, but these differences weren't tested one at a time.

Reproducing the results

On the jenna-testing branch, with the scripts in `analysis/headless/`:

```
node analysis/headless/sweep.js analysis/headless/sweeps/config_tick4452.json \
  --out analysis/headless/results/jenna-testing/config_tick4452

python analysis/headless/plot_evo_suicide.py analysis/headless/results/jenna-testing/config_tick4452 \
  --browser-view "soil cost 10" --browser-view "harmless burrowing"

python analysis/headless/plot_timeline.py analysis/headless/results/jenna-testing/config_tick4452
```

The 40 runs take about 13 minutes on an 8-core laptop. The per-run JSON files, the sweep file and the figures are in `analysis/headless/results/jenna-testing/config_tick4452/`.